

Blockchain Basics Cheat Sheet

12 tables · 90 concepts · Updated 2026-05-25 · cheatgrid.com

Blockchain is a distributed digital ledger that records transactions across many computers so the data is tamper-resistant and transparent. It combines **cryptography**, **consensus mechanisms** and **peer-to-peer networks** to create trustless systems where participants transact without intermediaries. The fundamentals, from how blocks link through cryptographic hashes to how a network reaches consensus, are what you need before working with cryptocurrencies, smart contracts or decentralized applications. The key insight: blockchain gets its security from mathematical proofs and network-wide verification, not from a central authority.

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TABLE 1

Core Components

The fundamental building blocks every blockchain shares — understanding how blocks, hashes, and transactions fit together gives you the mental model needed to reason about anything built on top.

Component	Example	Description
Block	Header: <code>prev_hash, merkle_root, timestamp, nonce</code> Body: <code>[tx1, tx2, tx3...]</code>	Container for transaction data consisting of a header (metadata) and body (transactions), linked via hashes to form the chain.
Transaction	<code>from: addr1, to: addr2, amount: 0.5 BTC, signature: ...</code>	Record of value transfer including sender, recipient, amount, and digital signature for authentication.
Hash	<code>SHA-256("hello") = 2cf24dba...</code> (64 hex chars)	- Cryptographic fingerprint of data — one-way function producing fixed-size output - any input change produces a completely different hash
Distributed Ledger	Every node stores full copy of blockchain	- Decentralized database replicated across all network nodes with no single point of control - all participants maintain identical records
Merkle Tree	<code>merkle_root = hash(hash(tx1+tx2) + hash(tx3+tx4))</code>	- Binary tree of transaction hashes enabling efficient verification of transaction inclusion without downloading the entire block - root stored in header
Nonce	<code>nonce = 2847563091</code>	- Random 32-bit number miners adjust to find a valid block hash - essential for the Proof-of-Work puzzle ("number used once").

Component	Example	Description
Genesis Block	Bitcoin Block 0: "Chancellor on brink..." Jan 3, 2009	• First block in a blockchain with no previous block reference • hardcoded into the protocol and establishes the chain's origin and initial state

TABLE 2

Consensus Mechanisms

Consensus mechanisms are how decentralized networks agree on the true state of the ledger without a central authority — the choice of mechanism directly determines the network's security, energy use, speed, and degree of decentralization.

Mechanism	Example	Description
Proof of Work (PoW)	Bitcoin mining: find <code>hash < target</code>	• Miners compete to solve a computational puzzle by finding a hash below the difficulty target • first to solve wins block reward • energy-intensive
Proof of Stake (PoS)	Ethereum: validators stake 32 ETH	• Validators selected based on staked cryptocurrency • probability proportional to stake • energy-efficient alternative to PoW • dishonest validators risk slashing
Delegated Proof of Stake (DPoS)	EOS: 21 elected block producers	• Token holders vote for a small group of delegates to validate blocks • high throughput and fast block times at the cost of some decentralization
Proof of Authority (PoA)	VeChain, private consortium chains	• Validators are pre-approved trusted nodes with known identities • very fast and low-overhead, suited for permissioned enterprise blockchains
Proof of History (PoH)	Solana: cryptographic timestamps embedded in ledger	• Cryptographic clock that encodes the passage of time directly into the blockchain, enabling validators to agree on transaction order without waiting for consensus rounds • enables Solana's high throughput
Byzantine Fault Tolerance (BFT)	Network functions with up to 33% malicious nodes	• System's ability to reach consensus despite faulty or malicious participants • ensures agreement even when nodes send conflicting information
Mining Difficulty	Adjusts every 2,016 blocks (≈2 weeks)	• Dynamic parameter controlling puzzle complexity • auto-adjusts to maintain target block time despite hashrate changes
Block Confirmation	1 conf = in block; 6 confs = highly secure	• Number of blocks added after a transaction's block • more confirmations = lower reversal risk • standard is 1–6 depending on value transferred
Validator	Stake 32 ETH → validate blocks → earn rewards	• PoS node that proposes and attests to blocks • selected based on stake • replaces miners in proof-of-stake systems
Finality	Transaction irreversible after 6 confirmations	• Point when a transaction becomes immutable and irreversible • varies by consensus type (instant in BFT, probabilistic in PoW, epoch-based in Ethereum PoS).
51% Attack	Attacker controls >50% hashrate → double-spend	• Occurs when an entity controls majority of network power, enabling transaction reversal and double-spending • economically infeasible on major chains

TABLE 3

Cryptography Fundamentals

Blockchain security rests entirely on cryptographic primitives — public-key cryptography, hash functions, and digital signatures make trustless verification possible without any central authority.

Technique	Example	Description
Private Key	e9873d79... (32 bytes, 256 bits) Never share!	- Secret key used to sign transactions and prove ownership - must be kept secure — losing it means permanent loss of access to funds
Public Key	04a1b2c3... (65 bytes uncompressed)	- Cryptographic key shared publicly to receive funds - mathematically derived from private key - safe to distribute openly
Digital Signature	ECDSA(privkey, tx_hash) → (r, s)	- Cryptographic proof that a transaction was authorized by the key owner - verifiable with public key - ensures authenticity and non-repudiation
Address	Bitcoin: 1A1zP1eP... (Base58) Ethereum: 0x742d35... (hex)	- Shortened hash of public key used as a receiving identifier - human-shareable format - derived via HASH160(pubkey) or similar
SHA-256	Input → a591a6d4... (always 256 bits)	- Cryptographic hash function producing a fixed 256-bit output - collision-resistant - core of Bitcoin's Proof-of-Work and block linking
ECDSA	Elliptic Curve Digital Signature Algorithm	- Standard signature scheme using elliptic curve cryptography - provides same security as RSA with smaller keys - used in Bitcoin and Ethereum
Zero-Knowledge Proof (ZKP)	Prover proves x satisfies a condition without revealing x	- Method of proving the validity of a statement without revealing the statement itself - enables privacy and efficient verification - foundational to ZK rollups
zk-SNARK	Zcash shielded transactions; zkSync Era proofs	- Zero-Knowledge Succinct Non-interactive Argument of Knowledge — small proof size, fast verification, but requires a trusted setup ceremony - used in most ZK rollups
zk-STARK	Starknet proofs (no trusted setup)	Zero-Knowledge Scalable Transparent Argument of Knowledge — no trusted setup required, post-quantum secure (hash-based), but produces larger proofs than SNARKs.

TABLE 4

Network Architecture

The peer-to-peer network layer determines how resilient and decentralized a blockchain actually is — node type, propagation speed, and access model all shape what the network can withstand.

Type	Example	Description
Peer-to-Peer (P2P)	Each node connects to 8–125 peers	- Decentralized network where nodes communicate directly without a central server - all participants are equal - enhances resilience
Full Node	Downloads and validates entire blockchain (500+ GB)	- Stores complete blockchain history and validates all transactions and blocks - enforces consensus rules independently - most secure node type
Mempool	Pending txs: [tx1: 50 sat/byte, tx2: 20 sat/byte...]	- Temporary storage for unconfirmed transactions awaiting inclusion in a block - each node maintains its own mempool - higher fees = faster inclusion

Type	Example	Description
Light Node (SPV)	Downloads only block headers (~80 bytes each)	• Stores block headers only • verifies transactions via Merkle proofs • lower resource requirements but partially trusts full nodes
Miner Node	Full node + mining software (PoW)	• Full node that attempts to create new blocks by solving Proof-of-Work puzzle • earns block rewards plus transaction fees
Block Propagation	New block → relay to peers → network-wide in seconds	• Process of broadcasting a new block across the network • propagation delay affects orphan block rate • optimized via compact block relay
Permissionless	Bitcoin, Ethereum: anyone can join	• Network where participation is open to all without approval • maximizes decentralization • characteristic of public blockchains
Permissioned	Hyperledger Fabric: invited members only	• Network requiring authorization to join and transact • controlled access suited for enterprises • trades decentralization for privacy and efficiency

TABLE 5

Transaction Models

How a blockchain accounts for balances and tracks value fundamentally shapes smart contract design, privacy properties, and how fees are calculated.

Model	Example	Description
UTXO (Unspent Transaction Output)	Bitcoin: inputs consumed → new outputs created	• Tracks discrete spendable outputs like digital cash • inputs are fully consumed to create new outputs • enhances parallelism and privacy
Account-Based	Ethereum: <code>balance[addr] += amount</code>	• Maintains a running balance per address like a bank account • simpler for smart contracts • easier to track total balance
Gas Fee	Ethereum: <code>(Base + Priority) × Gas Limit</code>	• Transaction cost paid to compensate validators • measured in gwei • higher fee = faster processing • base fee is burned (EIP-1559).
Transaction Pool	Mempool with priority queue by fee	• Collection of broadcast but unconfirmed transactions • miners/validators select from pool for the next block using fee-based prioritization
Double-Spending	Attempting to spend same BTC twice	• Attack where same funds are used in multiple transactions • prevented by consensus requiring acceptance of the longest/heaviest chain

TABLE 6

Block Structure

Every block in a blockchain follows a precise binary structure — understanding the individual fields explains how blocks are verified, linked, and mined.

Field	Example	Description
Block Header	Version, prev_hash, merkle_root, timestamp, bits, nonce (80 bytes)	• Compact 80-byte metadata section containing version, previous hash, Merkle root, timestamp, difficulty target, and nonce • the only part hashed in PoW
Previous Block Hash	0000000000000000000000008a89...	• Hash of the prior block's header • creates immutable chain linkage — changing any past block invalidates all subsequent blocks

Field	Example	Description
Block Reward	Bitcoin: 3.125 BTC (post-2024 halving)	• Newly minted cryptocurrency awarded to the block creator • combined with transaction fees • halves periodically to control supply
Difficulty Target	Compact bits: 0x1703a30c → target value	• Threshold that the block hash must fall below • lower target = higher difficulty • adjusts to maintain consistent block time
Timestamp	Unix time: 1704723600	• Records approximate block creation time • used for difficulty adjustment • must be greater than the median of the last 11 blocks
Block Body	List: [coinbase_tx, tx1, tx2...]	• Contains actual transaction data • variable size depending on block capacity • validated via Merkle root in header
Block Size	Bitcoin: 1 MB limit; Ethereum: dynamic	• Maximum data capacity per block • affects throughput and decentralization — larger blocks reduce the number of nodes that can validate

TABLE 7

Security and Immutability

Blockchain's security model is adversarial by design — these concepts define how the system resists tampering, manipulation, and centralization pressure from both external attackers and insiders.

Concept	Example	Description
Immutability	Altering Block 100 invalidates Blocks 101–200,000	• Property where recorded data cannot be changed without detection • achieved through cryptographic hashing and chain linkage
Trustless	No need to trust counterparty or intermediary	• System where participants transact without trusting each other • trust is replaced by cryptographic proofs and consensus • math-based security
Decentralization	No single entity controls the network	• Distribution of authority across many independent nodes • eliminates single points of failure and censorship • core blockchain principle
MEV (Maximal Extractable Value)	Validator reorders txs to front-run a DEX trade	• Maximum value extractable from block production by reordering, including, or excluding transactions • source of front-running and sandwich attacks in DeFi
Slashing	Ethereum: validator double-signs → 1+ ETH burned	• Penalty mechanism that destroys part of a validator's staked funds for provably malicious behavior such as double-signing • aligns economic incentives with honesty
Transparency	All transactions publicly viewable on block explorer	• Property where all network activity is visible to participants • public blockchains allow anyone to audit • privacy preserved through pseudonymous addresses
Tamper-Proof	Modify tx → hash changes → chain breaks	• Design making unauthorized alterations immediately detectable • hash functions and consensus ensure any modification is evident to all nodes

TABLE 8

Wallets and Key Management

A wallet is really a key manager, not a coin container — the funds live on-chain while the wallet holds the cryptographic secrets that prove ownership and authorize spending.

Type	Example	Description
Non-Custodial Wallet	MetaMask, hardware wallet: user controls keys	- User fully controls private keys - maximum sovereignty and full responsibility for security - hardware or software options available
Custodial Wallet	Exchange wallet: Coinbase holds your keys	- Third party controls private keys on your behalf - convenient but "not your keys, not your coins" - easier for beginners
Cold Wallet	Ledger, Trezor (offline hardware)	- Wallet stored offline (hardware or paper) - maximum security for long-term storage - immune to online attacks - requires physical access to sign
Hot Wallet	MetaMask, Trust Wallet (online)	- Wallet connected to the internet - convenient for frequent transactions - higher security risk - suitable for small daily amounts
Seed Phrase	12–24 words: <code>witch collapse practice...</code>	- Mnemonic backup for recovering a wallet - generates all private keys deterministically (BIP-39 standard) - must be stored securely offline
HD Wallet (BIP-32/BIP-44)	Single seed → unlimited child keys via derivation path	- Hierarchical Deterministic wallet that derives an entire tree of key pairs from one seed - enables address reuse avoidance and easy backup with a single phrase
Multisig Wallet	2-of-3: requires 2 signatures from 3 keyholders	- Wallet requiring multiple private-key signatures to authorize a transaction - eliminates single point of failure - used for treasuries and joint accounts

TABLE 9

Smart Contracts & Protocols

Smart contracts are the programmable layer of blockchain — they eliminate intermediaries by encoding business logic directly in tamper-proof code. Understanding the standards, governance patterns, and upgrade mechanisms is essential for any practical blockchain work.

Concept	Example	Description
Smart Contract	<code>if (condition) { transfer(amount); }</code> on Ethereum	Self-executing code deployed on blockchain - automatically enforces rules when conditions are met - immutable once deployed - no intermediaries required
ERC-20	USDC, DAI, UNI — <code>transfer()</code> , <code>balanceOf()</code>	- Ethereum standard for fungible tokens - defines a common API (transfer, approve, allowance) enabling interoperability across wallets and DeFi protocols
ERC-721	CryptoPunks, Bored Apes: unique token IDs	- Ethereum standard for non-fungible tokens (NFTs) - each token has a unique ID and cannot be divided - used for digital art, collectibles, and real-world asset tokenization
ERC-1155	Web3 game: swords (fungible) + unique skins (NFTs)	Multi-token standard supporting both fungible and non-fungible tokens in a single contract - enables batch transfers and reduces gas costs for gaming assets

Concept	Example	Description
Oracle	Chainlink feeds real-world price data on-chain	- Service that brings external data onto blockchain - solves the oracle problem (blockchains cannot access off-chain data natively) - critical infrastructure for DeFi
DAO (Decentralized Autonomous Organization)	MakerDAO: MKR holders vote on protocol parameters	Smart-contract-governed organization with no central authority - rules encoded in code - decisions made via token-holder voting - treasury controlled by the group
NFT (Non-Fungible Token)	Unique digital artwork or in-game item with provable ownership	- Blockchain token that is one-of-a-kind and non-divisible - ownership and provenance are publicly verifiable on-chain - implemented via ERC-721 or ERC-1155.
Fork (Hard)	Bitcoin → Bitcoin Cash (2017)	Non-backward-compatible protocol change - creates a permanent chain split - all nodes must upgrade - can result in two separate cryptocurrencies
Fork (Soft)	SegWit activation (2017)	Backward-compatible upgrade - old nodes still function - tightens rather than loosens rules - no chain split required
Halving	Bitcoin reward: 50 → 25 → 12.5 → 6.25 → 3.125 BTC	- Periodic reduction of block reward by 50% - occurs every 210,000 blocks (≈4 years) - controls supply inflation - next Bitcoin halving ≈2028.

TABLE 10

Scaling Solutions

Blockchains face a trilemma between security, decentralization, and throughput — scaling solutions attack the throughput constraint by moving computation off the base layer while inheriting its security guarantees.

Solution	Example	Description
Layer 1	Bitcoin, Ethereum base blockchain	- The base blockchain protocol itself - handles consensus and final settlement - security foundation - limited throughput by design
Layer 2	Lightning Network, Arbitrum, zkSync Era	Secondary protocol built atop L1 - processes transactions off-chain then settles on L1 - increases throughput without sacrificing L1 security
ZK Rollup	zkSync Era, Starknet, Polygon zkEVM	- L2 that batches thousands of transactions off-chain and posts a cryptographic validity proof (zk-SNARK/zk-STARK) to L1 - enables instant finality and fast withdrawals
Optimistic Rollup	Arbitrum, Optimism, Base	- L2 that assumes transactions are valid and relies on fraud proofs during a ~7-day challenge window to catch invalid batches - simpler to build, wider EVM compatibility
Sharding	Ethereum roadmap: parallel execution shards	- Horizontal partitioning of the blockchain into parallel shards that process transactions simultaneously - multiplies base-layer throughput without L2 dependence
State Channel	Lightning Network payment channels	- Off-chain communication channel between parties where only opening and closing transactions are settled on-chain - enables near-instant, fee-free repeated interactions
Plasma	OMG Network (legacy)	- Early L2 framework using a child chain anchored to Ethereum via Merkle roots - largely superseded by rollups due to data availability challenges and slow withdrawals

TABLE 11

DeFi Primitives

Decentralized finance (DeFi) recreates financial services — trading, lending, earning yield — directly in smart contracts, eliminating custodians; knowing the core primitives is prerequisite for building or auditing any DeFi protocol.

Primitive	Example	Description
DeFi (Decentralized Finance)	Aave, Uniswap, Compound on Ethereum	- Umbrella term for financial services built on public blockchains with no intermediaries - includes lending, trading, derivatives, and insurance powered by smart contracts
AMM (Automated Market Maker)	Uniswap: $x * y = k$ constant product formula	- DEX mechanism that prices assets algorithmically using a liquidity pool rather than an order book - anyone can trade 24/7 against pooled assets
Liquidity Pool	ETH/USDC pool on Uniswap	- Smart contract holding pairs of tokens deposited by liquidity providers - enables AMM trading - LPs earn a share of swap fees but face impermanent loss risk
Staking	Ethereum: deposit 32 ETH → earn ~2.9% APR	- Locking tokens in a PoS protocol to participate in consensus and earn rewards - staked capital acts as a bond that can be slashed for misbehavior
Yield Farming	Deposit LP tokens → earn governance token rewards	- Strategy of providing liquidity to earn additional protocol tokens on top of swap fees - rewards can come from token inflation and protocol incentives
Flash Loan	Borrow 1M USDC → arbitrage → repay — all in one tx	- Uncollateralized loan that must be borrowed and repaid within a single transaction - unique to DeFi - used for arbitrage, liquidations, and collateral swaps
Restaking	EigenLayer: restake stETH to secure AVS protocols	- Reusing already-staked assets to secure additional services (AVSs) and earn extra rewards - introduces additional slashing conditions and systemic risk

TABLE 12

Cross-chain & Interoperability

With thousands of isolated blockchains, interoperability protocols are the connective tissue of Web3 — they allow assets and data to move between networks, but every bridge also introduces new trust assumptions and attack surface.

Concept	Example	Description
Cross-chain Bridge	Wormhole, Chainlink CCIP, Polygon Portal	- Protocol that locks assets on a source chain and mints equivalent tokens on a destination chain - most blockchain hacks involve bridges due to their concentrated value and complexity
Atomic Swap	BTC ↔ LTC swap using HTLC	- Trustless cross-chain exchange using Hash Time-Locked Contracts (HTLCs) - funds are locked until a secret is revealed, ensuring the swap either completes fully or reverts
Sidechain	Polygon PoS, RSK (Bitcoin sidechain)	- Independent blockchain connected to a parent chain via a two-way peg - uses its own consensus rules - assets locked on the main chain are represented equivalently on the sidechain
Wrapped Token	WBTC: Bitcoin locked → ERC-20 token on Ethereum	- Token that represents an asset from another chain - the original is locked in a custodian or smart contract while a 1:1 pegged token is minted on the destination chain
IBC (Inter-Blockchain Communication)	Cosmos chains trading assets via IBC protocol	- Standardized protocol for trustless messaging and token transfers between independent blockchains - core interoperability layer of the Cosmos ecosystem - maintained as a formal spec

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